## * Normalization:

Normalization is a techinique often applied as part of data preparation of ML. The goal of normalisation is to change the values of numeric columns in a dataset to use a common scale, without distorting difference in the ranges of values or losing info.

techniques
- MinMax scaling
- mean normalization
- Max absolute
- Robust Scaling

### • MinMax Scaling:

lets take example of weights

| (xi) Weights | (xi) Normalize |
|---|---|
| 130 Kg | |
| 60 Kg | |
| 90 Kg → | |
| 47 Kg | |
| 72 Kg | |

now to normalize the given column we use,

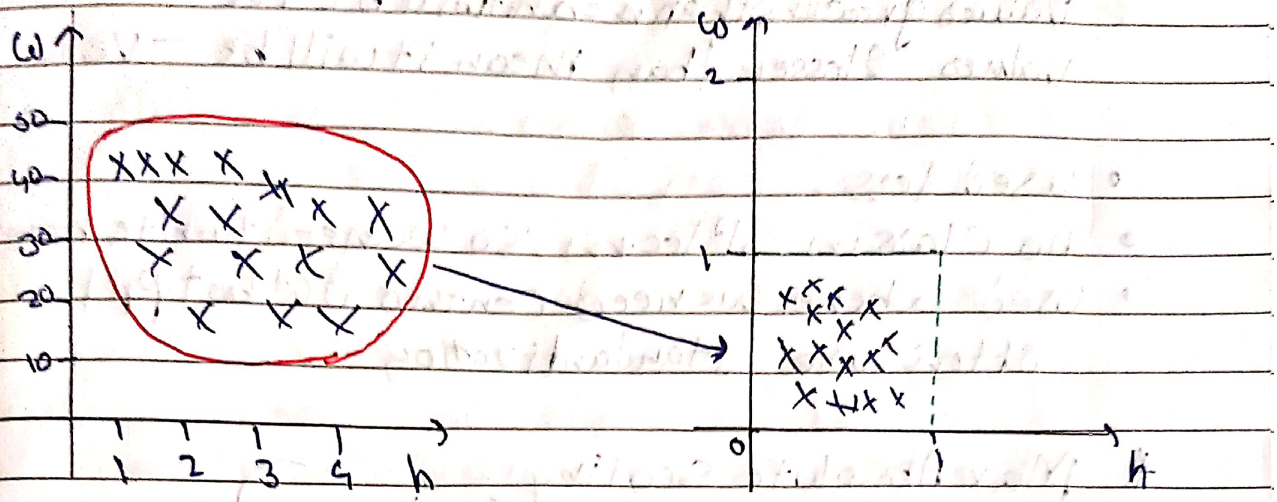
~~$X_i = X_i - x_{min}$~~

$$x_i = \frac{X_i - x_{min}}{x_{max} - x_{min}}$$

Its scale is always

[0 to 1]

geometry




Before Normalization

After Normalization

So as we can see that the data which was spread across in before in after it is squished in range of (0-1) in x, y axis (if it was in 3 dimension then it was squished in a cube and so on for other dimension

* ~~Min max~~ Mean normalization:

Weight (xi)

200

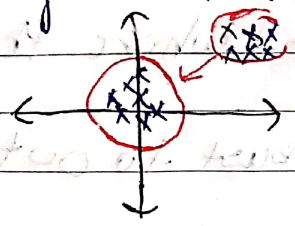
400

360

500

normalization $\rightarrow$ $x'i = \frac{xi - x_{mean}}{x_{max} - x_{min}}$

this is a mean centring that means bringing mean in centre of graph



Range is (-1 to 1)

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Values greater than mean it will be +ve
Values lesser than mean it will be −ve

- used less
- no Class in Sklearn, so we need to write manual
- Used where we need centred data but ppl often use standardization ...

* Max Absolute Scaling:
We have numerical data (xi)
formula: $xi' = \frac{xi}{|Xmax|}$

- Used in sparse data (where there are many 0)
- Has MaxAbsScaler in sciKit learn module

* Robust Scaling:

We have numerical data (xi)

formula = $xi' = \frac{xi - Xmedian}{IQR \{75^{th} \text{ percentile} - 25^{th} \text{ percentile}\}}$
(Interquartile range)

- Robust scale module in sciKit learn
- Used where there is many outlier
- Robust to outlier